

Design and Fabrication of a Coconut Dehusking Machine for Local Coconut Farmers in Asturias, Cebu

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Abstract

The Philippines, as the world's second largest coconut producer, relies heavily on manual dehusking methods that are slow, labor-intensive, and hazardous. This study presents the design, fabrication, and performance evaluation of a low-cost coconut dehusking machine developed for small-scale coconut farmers in Barangay Owak, Asturias, Cebu. The machine employs a dual counter-rotating roller blade mechanism powered by a single-phase 1 hp electric motor coupled to a 1:60 speed reducer, achieving a controlled blade speed of 14.33 rpm. Design parameters were derived from physical measurements of 50 dry coconut samples (mean x-axis diameter: 17.53 cm; y-axis diameter: 20.57 cm; husk thickness: 19.5 mm). Machine performance was evaluated on a sample of 20 randomly selected dry coconuts. The mean dehusking time was 13.94 seconds per coconut (range: 11.5–15.9 s), meeting the target threshold of ≤ 15 seconds and yielding an estimated throughput of 258 coconuts per hour. The design prioritizes operator safety through a low blade rotational speed, cost-effectiveness through locally sourced components, and mechanical simplicity for ease of maintenance. Total fabrication cost was PHP 39,423.50 (approximately USD 690). Results confirm that the fabricated machine is a viable, safe, and affordable alternative to manual dehusking for rural agricultural communities in Asturias, Cebu.

Keywords: coconut dehusking; roller blade mechanism; counter-rotating rollers; agricultural machinery; small-scale farming; speed reducer; shear force; SolidWorks simulation

1. Introduction

The coconut palm (*Cocos nucifera* L.) is one of the most economically significant tropical crops in the world, cultivated across more than 90 countries with an annual global production exceeding 5.4 billion tons [7]. The Philippines, recognized as the world's second largest coconut producer, dedicates approximately 3.26 million hectares — roughly one-third of its total arable land — to coconut cultivation across 68 of its 79 provinces. The industry sustains the livelihoods of about one-third of the Filipino population and serves as the country's primary agricultural export [1,2].

In rural communities such as Barangay Owak, Asturias, Cebu, coconut dehusking is central to local livelihoods, particularly in the production of bibingka, a traditional Filipino delicacy relying on coconut as a primary ingredient. Conventionally, dehusking is performed manually using a machete or a locally crafted steel spike called a bolo: the worker cuts the stem and apical ends, makes longitudinal incisions, and levers off the husk [9]. While skilled workers can dehusk up to 1,080 coconuts in a six-hour shift (~3 nuts/minute) [10], manual dehusking is physically demanding, inconsistent in speed due to fatigue, and carries significant injury risk from sharp tools [4].

Studies report that the efficiency of manual dehusking ranges between 64.55% and 82%, compared to efficiency values of 90.42%–93.45% achieved by motorized machines [5,6,11]. Despite the performance advantage of mechanized solutions, many existing designs — employing pneumatic systems, hydraulic clamps, or high-speed rollers — are financially inaccessible and technically complex for small-scale farmers in remote areas. High

fabrication costs, dependence on specialized components, and maintenance requirements beyond the skill of non-specialist operators limit their adoption in communities like Asturias, Cebu [8,20].

This study addresses that gap by designing, fabricating, and evaluating a low-cost, mechanically simple coconut dehusking machine using widely available local components. Three specific objectives guided the work: (1) to design and fabricate the machine tailored to local farming conditions; (2) to determine the appropriate power and speed specifications targeting a dehusking time of ≤ 15 seconds per coconut and a capacity of up to 276 nuts per hour; and (3) to evaluate the actual dehusking speed and capacity of the fabricated prototype through empirical testing.

2. Literature Review

2.1 Coconut Physical Properties and Dehusking Force

Effective machine design requires an understanding of the physical and mechanical properties of coconut. Varghese et al. [13,14] emphasize that size, shape, weight, shell diameter, and husk thickness are critical parameters for machine design. From a sample of 35 dry coconuts, Varghese [14] recorded an average shell thickness of 3.00 mm, shell diameter of 95.54 mm, and a shell rupture force of 4,427.10 N. The husk separating force — more relevant for dehusking machines — was reported at approximately 865.7 N for dry coconuts with an average moisture content of 1.3%. Sakhare et al. [31] corroborated this using FEM analysis (ANSYS 16.0), finding that a peeling force of 114 N produces shear stresses in the range of $\pm 1.77 \times 10^6$ Pa in structural steel blades.

2.2 Existing Dehusking Machine Designs

Spike roller machines use inward-facing spiked rollers rotating at different speeds to grip and detach the husk [5]. Ovat and Odey [18] added a screw conveyor between counter-rotating spiked rollers to enable auto-feeding, achieving 92.5% efficiency at 120 coconuts per hour. **Clamp/impaling** methods use a fixed spear or hydraulic claw to pierce and strip the husk [21], but carry inherent safety risks from exposed mechanisms. **Pneumatic roller/conveyor** systems, exemplified by Alcantara et al. [20], combine automated feeding, opposite-direction spiked rollers, and an integrated cutter to achieve the highest productivity while minimizing worker fatigue. However, pneumatic system complexity and cost make these unsuitable for small-scale operators.

For power transmission, reviewed designs employ worm gear drives (Azmi et al. [11], 70:1 ratio, 21 rpm output), belt drives for smooth transmission, chain-sprocket systems for adjustable ratios, and spur gears for compact reduction [3,24,37]. Selvan and Das [19] demonstrated that a simpler 1 hp, 3-phase motor with gear reducer can dehusk a coconut in 14 seconds at 220 nuts per hour. A single-phase 2 hp induction motor at 1500 rpm used by Azmi [11] achieved 90.42% efficiency at 222 nuts per hour.

2.3 Research Gap

While existing designs have advanced mechanized dehusking, a consistent gap remains: most optimized machines incorporate high-speed rollers (>60 rpm) or complex sub-systems (pneumatics, hydraulics, screw conveyors) that are financially and technically inaccessible to small-scale farmers in rural Philippine communities. Three specific unmet needs were identified: (1) a design using only basic, locally available mechanical components (V-belts, spur gears, roller chains) at low fabrication cost; (2) a low blade rotational speed (≤ 30 rpm) for improved operator safety during manual coconut feeding; and (3) a machine dimensioned using empirically measured local coconut physical properties. The present study directly addresses all three gaps.

3. Methodology

3.1 Research Environment and Coconut Sample Characterization

The study was conducted in Barangay Owak, Asturias, a third-class municipality in Cebu Province, Philippines, where coconut farming and coconut-based food production are primary livelihoods. A sample of 50 dry coconuts was randomly selected to measure physical properties. Dry coconuts — harvested 3–4 months after propagation and air-dried for 8–10 days — were chosen because they represent the type most commonly processed by local farmers and because their lower moisture content ($\sim 1.3\%$) reduces the required husk-separating force compared to fresh green coconuts. Measurements were taken of the x-axis diameter (longitudinal), y-axis diameter (latitudinal), and husk thickness for each sample. These data established the design parameters for the roller geometry and blade length.

Physical Property	Sample Size	Mean Value	Design Parameter Used
Diameter — x-axis (longitudinal)	50	17.53 cm	Roller blade length: 430 mm
Diameter — y-axis (latitudinal)	50	20.57 cm	Roller gap and feed width
Husk thickness	50	19.5 mm	Blade width: 20 mm

Table 1. Measured physical properties of 50 dry coconut samples and derived design parameters.

3.2 Machine Design Concept

The dehusking mechanism is based on a dual counter-rotating roller blade system. Two parallel rollers, each fitted with six radially arranged flat blades (20 mm × 400 mm × 6 mm, structural steel), rotate inward toward each other. A coconut is manually fed longitudinally — parallel to the roller axis — between the two rollers from above. As the blades engage the husk, the counter-rotation applies shear and tensile forces to the fibrous mesocarp, progressively stripping it from the inner shell. Longitudinal feeding orientation, validated by Azhikodan [26] to reduce dehusking duration, was adopted as a core design principle.

Each roller drum consists of a hollow shaft (outer diameter: 60 mm, inner diameter: 54 mm) with six blades welded at equal angular spacing. The moment arm for torque calculation is $r_a = 30 \text{ mm}$ (drum radius) + 20 mm (blade length) = 50 mm.

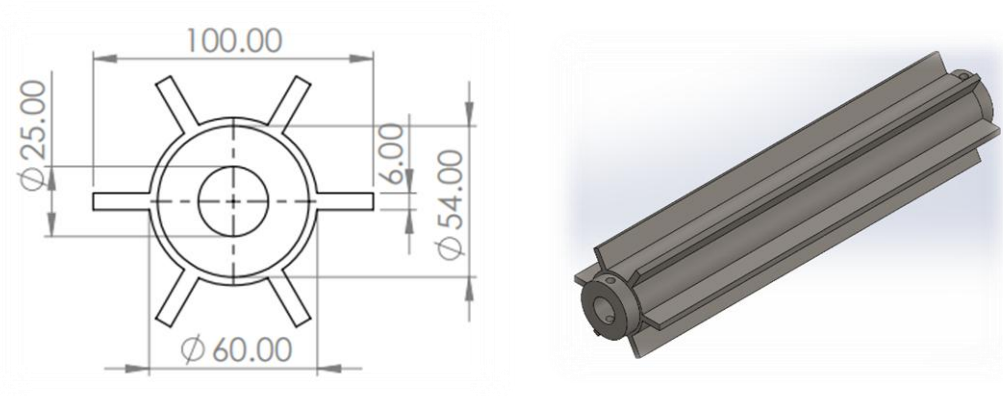


Figure 1. Roller blade profile & Roller blade (SolidWorks engineering drawing). Hollow shaft OD = 60 mm; ID = 54 mm; blade width = 20 mm; blade length = 430 mm. All dimensions in mm.

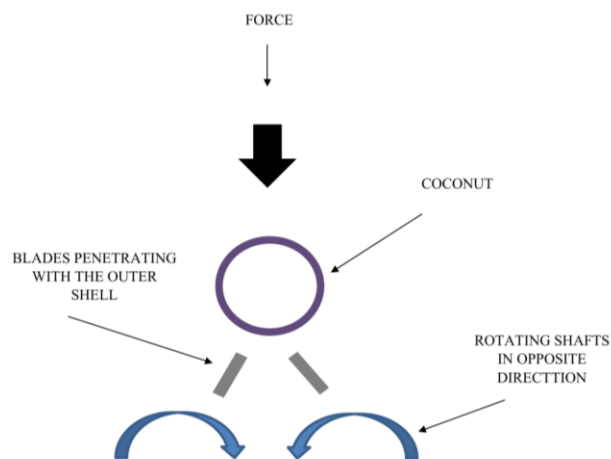


Figure 2. Principle of working: counter-rotating shafts apply downward force through blades penetrating the coconut husk, separating it from the inner shell.

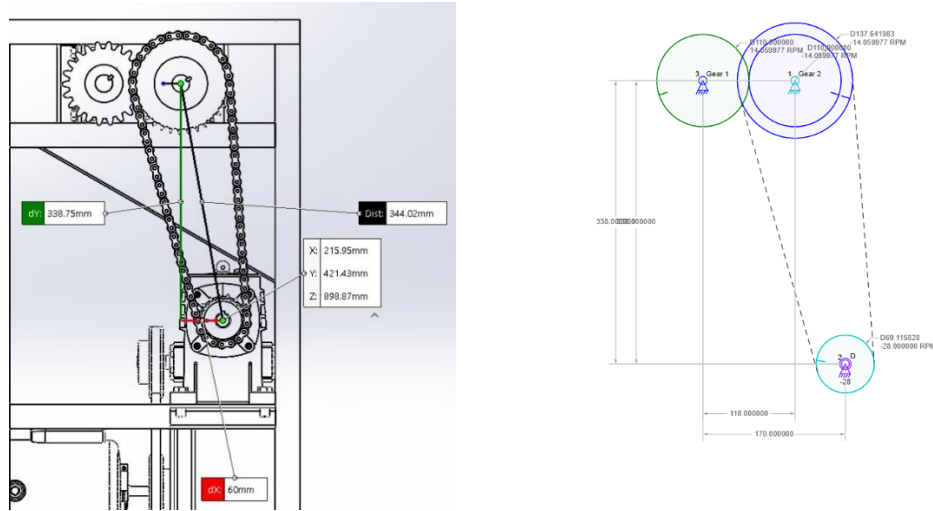


Figure 3. Speed Reducer to Roller Blades Connection

3.3 Power and Speed Calculations

The power requirement was determined from the husk-separating force of 865.7 N (Varghese [14]) and the required torque at the rollers. Using $P = 2\pi NT$:

Parameter	Symbol	Value
Husk separating force (per blade)	F_c	865.7 N
Moment arm (drum radius + blade width)	r_a	0.050 m
Torque per drum	T	43.285 N·m
Total torque (2 drums)	T_{total}	86.57 N·m
Target blade speed (2 nuts/min $\times$ 7 rev/nut)	N	14 rpm
Required power (6 blades)	P	≈ 1.03 hp
Selected motor	—	1 hp, single-phase, 1,720 rpm

Table 2. Power and speed calculation summary.

A standard 1 hp (0.746 kW) single-phase induction motor at 1,720 rpm was selected. The drive train achieves the target blade speed through three reduction stages:

Stage	Component	Input Speed	Output Speed	Ratio
1	Motor $\rightarrow$ V-belt $\rightarrow$ Speed Reducer	1,720 rpm	28.67 rpm	1:60
2	Speed Reducer $\rightarrow$ Chain-Sprocket (17T driver, 34T driven)	28.67 rpm	14.33 rpm	1:2
3	Sprocket $\rightarrow$ Spur Gears $\rightarrow$ Roller Blades	14.33 rpm	14.33 rpm*	1:1

Table 3. Drive train reduction stages. *Spur gears reverse direction to opposing roller; speed magnitude equal, direction opposite.

3.4 Structural Design and Material Selection

The machine frame was constructed from $45 \times 45 \times 5$ mm angle bars. Roller blades and shafts were fabricated from medium-carbon structural steel (C: 0.25–0.55%), selected for its balance of strength, machinability, and cost. Material design was validated using the maximum shear design stress criterion (Khurmi [37]):

$$S_{sd} = 0.18 \times S_{ut} = 0.18 \times 670 \text{ MPa} = \mathbf{120.6 \text{ MPa}}$$

Von Mises stress distribution across critical shaft cross-sections was evaluated through SolidWorks Simulation (FEA), confirming that principal stresses remain below the allowable limit under combined bending and torsional loads at 14.33 rpm. Four UCP 205-16 pillow block bearings support the two roller shafts. The complete bill of materials is presented in Table 4.

Component	Specification	Unit Cost (PHP)	Total (PHP)
AC Motor	1 hp single-phase induction motor	4,412.00	4,412.00
Roller Blades (×2)	6 blades each; custom fabricated	3,300.00	6,600.00
Speed Reducer	1:60 ratio, Body 60	5,500.00	5,500.00
Helical Gears (×2)	Surplus	500.00	1,000.00
Machining — Bore/Build	Cable Hydrocontrol (×4 ops)	625.00	2,500.00
Sprocket set	40T large / 16T small	3,800.00	3,800.00
V-Belt (Type A)	A39 Bando Classic, 1041.4 mm	300.00	300.00
Pulley	Standard	300.00	300.00
Solid Shafts (×2)	Sourced from junkshop	30.00/kg	355.00
Keyway & Shaft Alignment	Cable Hydrocontrol	—	3,300.00
Pillow Block Bearings (×4)	UCP 205-16	350.00	1,400.00
Angle Bar Frame	45×45×5 mm, 3 m total	1,910.00	1,910.00
Bolts and Screws	Assorted	—	639.00
Electrical components	3 m wire, plug, circuit breaker	—	507.50
Fabrication & Assembly	Cable Hydrocontrol labor	—	6,900.00
TOTAL			39,423.50

Table 4. Bill of Materials — complete cost breakdown (PHP, June 2024).

3.5 Fabrication Procedure

Fabrication was carried out at Cable Hydrocontrol Machining Center, Cebu. The assembly sequence proceeded as follows: (1) frame construction from angle bars by welding; (2) shaft turning and keyway cutting on a lathe; (3) roller blade welding onto hollow shaft at 60° angular intervals; (4) bearing and pillow block mounting on frame; (5) spur gear and sprocket press-fit and key installation; (6) chain assembly and V-belt installation; (7) motor and speed reducer mounting and alignment; (8) electrical wiring (motor, breaker, plug). SolidWorks 3D assembly modeling was used throughout to verify dimensional fit and detect interference before physical fabrication.

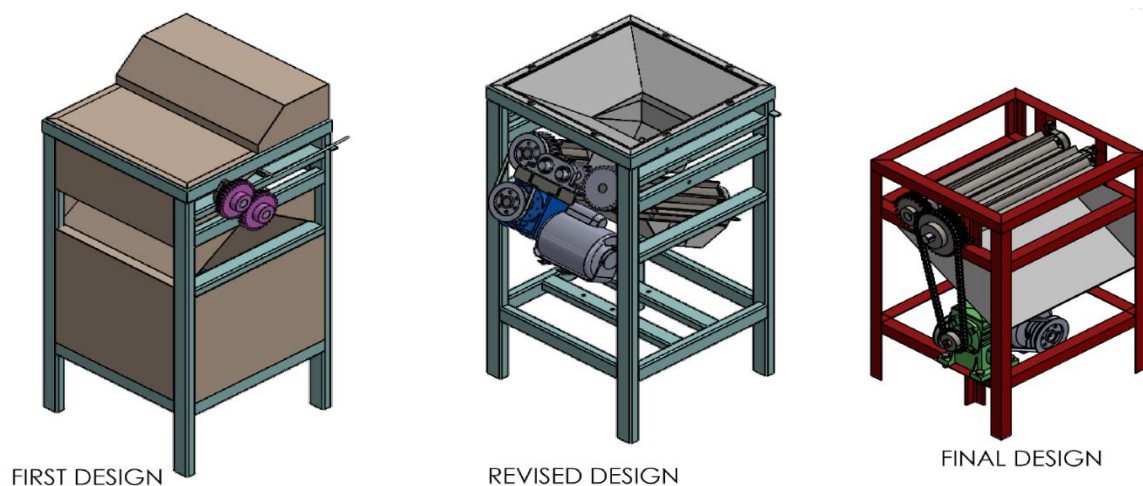


Figure 4. Prototype development progression rendered in SolidWorks: first design concept (left), revised design with exposed drive components (middle), and finalized prototype (right).

Materials (Hollow Tube and Steel Plates)



Roller Blades and Solid Shaft



Figure 5. Fabrication documentation: raw materials (hollow tube and steel plates, upper); fabricated roller blades and solid shaft components (lower).



Figure 6. Machine assembly: frame with roller and chain-sprocket drive system (upper); motor and speed reducer mounting in progress (lower).

3.6 Performance Testing Protocol

Performance evaluation was conducted on a random sample of 20 dry coconuts. Each coconut was manually fed longitudinally between the rollers with slight guiding pressure until fully dehusked. Dehusking time was recorded from first blade contact to complete husk removal using a stopwatch. Descriptive statistics were applied: mean, minimum, maximum, range, and frequency distribution across three equal intervals. The primary success criterion was mean dehusking time ≤ 15 seconds per coconut, equivalent to the benchmark manual dehusking rate.

4. Results

Table 5 presents the dehusking time recorded for each of the 20 test coconuts. Times ranged from a minimum of **11.5 seconds** to a maximum of **15.9 seconds**, with a mean of **13.94 seconds per coconut** and a range of 4.4 seconds.

Coconut No.	Time (s)	Coconut No.	Time (s)	Coconut No.	Time (s)	Coconut No.	Time (s)
1	14.4	6	14.8	11	13.6	16	15.0
2	12.8	7	11.5	12	13.0	17	12.5
3	13.3	8	14.8	13	12.1	18	11.8
4	13.4	9	14.7	14	15.8	19	14.9
5	13.1	10	11.8	15	15.9	20	13.3
Mean	13.94 s	Min	11.5 s	Max	15.9 s	Range	4.4 s

Table 5. Dehusking speed results for 20 dry coconut samples (seconds/coconut).

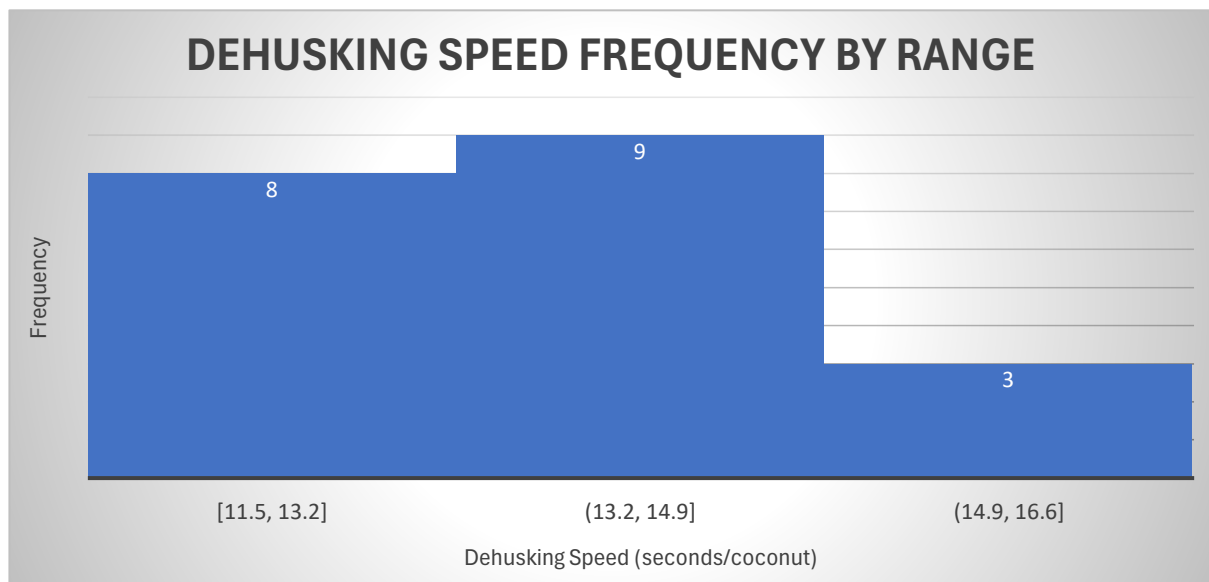


Figure 7. Frequency distribution of dehusking times across the 20-coconut sample. Modal range: 13.2–14.9 s (9 instances, 45%); 11.5–13.2 s (8 instances, 40%); 14.9–16.6 s (3 instances, 15%).

The frequency distribution (Figure 6) reveals that **85% of dehusking events** (17 of 20 coconuts) occurred between 11.5 and 14.9 seconds, with the modal range at 13.2–14.9 seconds (9 instances, 45%). Only 3 coconuts (15%) required more than 14.9 seconds, with the maximum of 15.9 seconds still within an acceptable margin above the 15-second target. The mean dehusking time of 13.94 s/coconut translates to an estimated throughput of **4.3 coconuts per minute** or **approximately 258 coconuts per hour** under continuous operation — approaching the 276 nuts/hour design target.

5. Discussion

5.1 Performance Against Design Objectives

All three research objectives were met. The fabricated machine achieved a mean dehusking time of 13.94 s/coconut — within the ≤ 15 s design target and well within twice the manual rate (≤ 30 s). The machine met the throughput objective of 2–5 nuts per minute, and the confirmed blade speed of 14.33 rpm satisfies the ≤ 30 rpm safety threshold. These results align with comparable literature findings: Selvan and Das [19] reported 14 s/coconut and 220 nuts/hour with a 1 hp motor, while Azmi [11] achieved 222 nuts/hour with a 2 hp motor. The present design achieves competitive throughput at half the motor power requirement.

5.2 Comparison with Existing Designs

Study	Motor	Speed (rpm)	Time (s/nut)	Capacity (nuts/hr)	Drive System
Azmi et al. [11]	2 hp, 1-phase	21	—	222	Worm gear + V-belt
Selvan & Das [19]	1 hp, 3-phase	50	14	220	Gear reducer + rollers
Alcantara et al. [20]	Not stated	Variable	—	High*	Pneumatic rollers + conveyor
Ovat & Odey [18]	Not stated	—	—	120	Counter-rotating spiked rollers
Present study	1 hp, 1-phase	14.33	13.94	~258	V-belt → speed reducer → chain-sprocket → spur gears

Table 6. Comparative performance of selected coconut dehusking machine designs. *Alcantara et al. reported the highest productivity but did not provide a specific nuts/hour figure.

The present design achieves throughput comparable to higher-powered or more complex machines, while using a 1 hp single-phase motor — the lowest power specification among comparable roller-type designs. The significantly lower blade speed (14.33 rpm vs. 50+ rpm in most comparable designs) is the key safety differentiator: slower blades allow the operator to correct positioning or remove a misaligned coconut without injury risk from fast-moving components.

5.3 Cost and Accessibility

The total fabrication cost of PHP 39,423.50 (~USD 690 at June 2024 rates) was achieved through deliberate component selection: sourcing shafts from local junkshops, using a surplus gear, and utilizing widely available A39 V-belt and standard sprocket sizes. All major components were sourced from Cebu City suppliers, ensuring local replaceability and straightforward maintenance. The absence of pneumatic or hydraulic components eliminates the most common barrier to small-farm adoption.

5.4 Limitations

Several limitations should be acknowledged. First, the performance sample was limited to 20 coconuts, which constrains statistical power; a larger sample would provide more robust estimates of mean throughput and variability. Second, all tested coconuts were sourced from Asturias and may not represent the full range of coconut sizes and husk characteristics across other regions. Third, long-duration operational testing (fatigue, wear, thermal behavior) was not conducted, and the machine's durability over extended field use remains uncharacterized. Fourth, the study did not measure husk completeness (proportion of husk successfully removed per run) as a quantitative metric. Fifth, socioeconomic factors — farmer adoption barriers, financing availability, and cultural acceptance — were beyond the scope of this study but are critical for real-world deployment.

6. Conclusion

This study successfully designed, fabricated, and evaluated a low-cost coconut dehusking machine for small-scale coconut farmers in Asturias, Cebu. The machine employs a dual counter-rotating roller blade mechanism driven by a 1 hp single-phase motor through a cascaded drive train (V-belt → 1:60 speed reducer → 1:2 chain-sprocket → spur gears), delivering a blade speed of 14.33 rpm. Empirical testing on 20 dry coconuts produced a mean dehusking time of 13.94 seconds per coconut — meeting the ≤ 15 s design criterion — with an estimated throughput of 258 coconuts per hour.

The blade speed of 14.33 rpm satisfies the ≤ 30 rpm safety requirement, substantially reducing operator injury risk compared to higher-speed designs in literature. The total fabrication cost of PHP 39,423.50 (~USD 690), achieved through locally available components and community-based machining services, represents a financially accessible option for small-scale farmers. All design objectives were met, and the machine demonstrates performance competitive with higher-powered and more complex published designs while maintaining mechanical simplicity.

The findings contribute to the growing body of work on low-cost agricultural mechanization for developing-economy farming communities, and the design serves as a scalable, maintainable baseline for future development in Philippine coconut farming regions.

7. Recommendations for Future Research

Based on the study's findings and acknowledged limitations, the following are recommended:

1. **Expanded performance trials.** Conduct testing on $n \geq 100$ coconuts from multiple moisture content levels and size categories to generate statistically robust performance data, including husk completeness as a quantitative metric.
2. **Blade profile optimization.** Investigate curved, sharper-tipped blade profiles (30° tip angle, R12.5 mm radius, as recommended by the thesis panel) to improve initial husk penetration and reduce time variability (Figure 7).
3. **Idling mechanism.** Add a clutch or idling mechanism to allow continuous motor operation without frequent start-stop cycling, reducing motor wear and energy consumption.
4. **Moisture content study.** Systematically determine the optimal coconut dryness level (harvest age and post-harvest drying duration) that minimizes husk-separating force and dehusking time.
5. **Durability and wear testing.** Conduct extended operational trials (≥ 500 hours) to characterize blade wear, bearing fatigue, and chain elongation over time.
6. **Socioeconomic adoption study.** Assess farmer willingness to adopt, financing mechanisms, and the economic impact on productivity and income in Asturias, Cebu.
7. **Smart integration.** Explore integration of sensors for automatic coconut detection and feed speed control, building toward low-cost semi-automated operation.

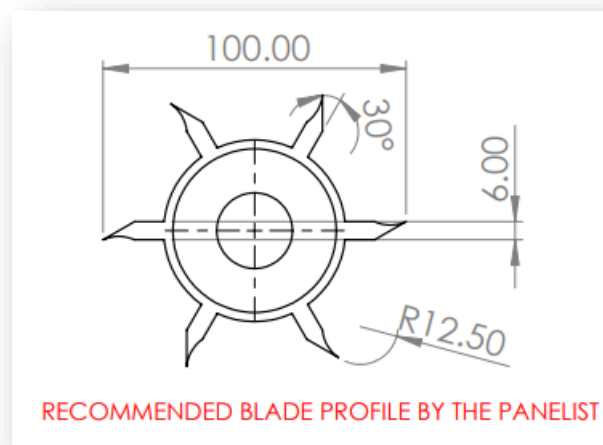


Figure 8. Recommended blade profile by the thesis evaluation panel: curved tips at 30° angle with R12.5 mm radius for improved husk penetration performance.

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